Report: Large N Quantum Anomalous Metal

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Abstract

Based on the semi-magnetic weak topological insulator (SMWTI) film structure, we find $N = 2$ quantum anomalous metal phase with unit quantum Hall conductance inside the emergent parity invariant regimes as another realization of the metallic quantum anomalous Hall effect (MQAHE) without chiral edge states. We also prove the absence of $N = 4$ phase in semi-magnetic TI as a consequence of Fermion-doubling. However, chance of realizing $N = 2 + 2$ phase still exists, providing that an effectively staggered potential is added.

I. Realization of $N = 2$ Phase in Semi-Magnetic Weak Topological Insulator Film

The Hamiltonian of 3-D **bulk** TI[1–5] is represented by the standard modified Dirac's equation[6] on lattice $\mathcal{H}_{TI} = \sum_{\boldsymbol{k}} \Psi^{\dagger}_{\boldsymbol{k}} H(\boldsymbol{k}) \Psi_{\boldsymbol{k}}$ where

$$H(\boldsymbol{k}) = \sum_{\mu} \lambda_{\mu} \sin(k_{\mu} a_{\mu}) \alpha_{\mu} + \left[m_0 - 4 t_{\mu} \sin^2 \left(\frac{k_{\mu} a_{\mu}}{2} \right) \right] \beta, \tag{1}$$

and given that $\lambda_{\mu} > 0$ and $t_{\mu} > 0$, the topology of the system is determined by the low-energy mass term m_0, which drives the system into trivial, weak TI or strong TI with four indices.

An anisotropic 3-D topological insulator **film** with limited number of sites along z direction, by Fourier transformation in $x - y$ plane, reads

$$\mathcal{H}_{\text{Film}} = \sum_{l_z, \boldsymbol{k}} \left(\Psi^{\dagger}_{l_z, \boldsymbol{k}} \mathcal{M}_0(\boldsymbol{k}) \Psi_{l_z, \boldsymbol{k}} + \Psi^{\dagger}_{l_z, \boldsymbol{k}} \mathcal{T}_z \Psi_{l_z+1, \boldsymbol{k}} + \text{h.c.} \right) + \sum_{l_z, \boldsymbol{k}} \Psi^{\dagger}_{l_z, \boldsymbol{k}} H_{\parallel} \Psi_{l_z, \boldsymbol{k}}, \tag{2}$$

with

$$H_{\parallel} = \lambda_{\parallel} [\sin(k_x a) \sigma_x \tau_x + \sin(k_y b) \sigma_y \tau_x], \tag{3}$$

and $\mathcal{M}_0(\boldsymbol{k}) = M_0(\boldsymbol{k}) \sigma_0 \tau_z = [m_0(\boldsymbol{k}) - 2 t_{\perp}] \sigma_0 \tau_z$, where

$$m_0(\boldsymbol{k}) = m_0 - 4 t_{\parallel} \left(\sin^2 \frac{k_x a}{2} + \sin^2 \frac{k_y b}{2} \right). \tag{4}$$

Note that we have set $t_x = t_y = t_{\parallel}$, $t_z = t_{\perp}$, $\lambda_x = \lambda_y = \lambda_{\parallel}$, $\lambda_z = \lambda_{\perp}, a = b$.

The **equivalently effective model**[7, 8] is obtained by:

(i) Separate Hamiltonian at $\boldsymbol{k}$ as

$$\mathcal{H}_{\text{Film}}(\boldsymbol{k}) = \mathcal{H}_{1d}(\boldsymbol{k}) + \mathcal{H}_S(\boldsymbol{k}) \tag{5}$$

with

$$\mathcal{H}_{1d}(\boldsymbol{k}) = \sum_{l_z} \left(\Psi^\dagger_{l_z,\boldsymbol{k}} \mathcal{M}_0(\boldsymbol{k}) \Psi_{l_z,\boldsymbol{k}} + \Psi^\dagger_{l_z,\boldsymbol{k}} \mathcal{T}_z \Psi_{l_z+1,\boldsymbol{k}} + \text{h.c.} \right), \tag{6a}$$

$$\mathcal{H}_S(\boldsymbol{k}) = \sum_{l_z} \Psi^\dagger_{l_z,\boldsymbol{k}} H_\parallel \Psi_{l_z,\boldsymbol{k}}. \tag{6b}$$

(ii) Solve $\mathcal{H}_{1d}$ with eigenstates Φ^n_i whose energies are $(m_n, -m_n, -m_n, m_n)$ for $i = 1, 2, 3, 4$ and $n = 1, \cdots, L_z$. Notice that the sequence of basis has been assigned so that $i = 1, 2$ states and $i = 3, 4$ states are separated by mirror symmetry. This procedure is general, depending only on the specific mirror or parity symmetry the system owns and not on specific parameter choices.

(iii) Project the film Hamiltonian and possible external field terms using Φ^n_i as

$$(H_{\text{equ}})^{nn'}_{ij} = \langle \Phi^n_i | \, H_{\text{Film}} + V_{\text{ext}} \, | \Phi^{n'}_j \rangle \, . \tag{7}$$

Basically, the projection of the original film Hamiltonian leads to

$$\tilde{H}_{\text{Film}}(\boldsymbol{k}) = \bigoplus_{n=1}^{L_z} \left[\lambda_\parallel (\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + m_n(\boldsymbol{k})\tau_z\sigma_z \right], \tag{8}$$

where $2L_z$ Dirac fermions $\tilde{H}_{\text{Film}} = \oplus_{n,\chi} h_{n,\chi}(\boldsymbol{k})$ emerge as

$$h_{n,\chi}(\boldsymbol{k}) = \lambda_\parallel (\sin(k_x a)\sigma_x + \sin(k_y b)\sigma_y) + \chi m_n(\boldsymbol{k})\sigma_z, \tag{9}$$

with $\chi = \pm$ labelling the mirror eigenvalue.

Clearly, $\mathcal{M}_0(\boldsymbol{k})$ in $\mathcal{H}_{1d}(\boldsymbol{k})$ would be the main element dominating whether it has zero modes solution, under suitable boundary condition. In the numerical simulation, we set the model parameters to be fixed as $\lambda_\parallel = 0.41$eV eV, $\lambda_\perp = 0.44$eV, $t_\parallel = 0.566$eV, $t_\perp = 0.4$eV, $a = b = 1$nm, $c = 0.5$nm[4] **except** for the static low energy mass m_0.

Notice that the above solution owns its **continuum model** correspondence. And in the half quantum mirror Hall effect[8], we have proved that for the continuum model solution, the lowest mass term is

$$m_1^{\text{continuum}}(\boldsymbol{k}) = \Theta(-m_0(\boldsymbol{k}))m_0(\boldsymbol{k}), \tag{10}$$

where expression in the Heaviside theta function is the topological criterion for continuum model of 1-D TI[9, 10]

$$h_{1d}^{\text{continuum}}(\boldsymbol{k}) = -i\lambda_\perp \partial_z \tau_x + (m_0(\boldsymbol{k}) + t_\perp \partial_z^2)\tau_z. \tag{11}$$

Since we assume $t_\perp > 0$, the non-trivial case with zero-energy mode on the 1-D boundary as two end points is identified by $m_0(\boldsymbol{k}) > 0$ with $\Theta(-m_0(\boldsymbol{k})) = 0$, and the existence of this zero-mode exactly corresponds to the existence of surface states with Dirac like linear dispersion of 3-D TI film.

Pushing the claim above to **lattice model** and we see that the existence of surface states in TI film is determined by the existence of boundary zero-mode of the 1-D lattice TI Hamiltonian h_{1d}^{lattice}, which is obtained by taking quantum number $\sigma_z = 1$ in $\mathcal{H}_{1d}$, and is the discretized latticing version of $h_{1d}^{\text{continuum}}(\boldsymbol{k})$. Since from its bulk form

$$h_{1d}^{\text{lattice}}(\boldsymbol{k}, k_z) = \lambda_\perp \sin(k_z c)\tau_x + \left[m_0(\boldsymbol{k}) - 4t_\perp \sin^2\left(\frac{k_z c}{2}\right)\right]\tau_z, \tag{12}$$

we have the non-trivial condition

$$0 < m_0(\boldsymbol{k}) < 4t_\perp. \tag{13}$$

Corresponding, we have its lowest energy solution as

$$m_1(\boldsymbol{k}) = \Theta\left[-m_0(\boldsymbol{k})\right] m_0(\boldsymbol{k}) + \Theta\left[m_0(\boldsymbol{k}) - 4t_\perp\right](4t_\perp - m_0(\boldsymbol{k})). \tag{14}$$

Further considering the linear term leads to the criterion for the existence of surface states constrained by three sets

$$\begin{cases} 0 < m_0 < 4t_\perp \\ 4t_\parallel < m_0 < 4t_\parallel + 4t_\perp \\ 8t_\parallel < m_0 < 8t_\parallel + 4t_\perp \end{cases}, \tag{15}$$

corresponding to single Dirac cone at $\Gamma = (0,0)$, double Dirac cones at $X = (\pi, 0), Y = (0, \pi)$ and single Dirac cone at $M = (\pi, \pi)$, respectively. The first and third cases lead to strong TI with single Dirac cone, while the second case provides with two Dirac cones connected, which shall be the case we are interested in here. Without losing generality, we choose

$$m_0 = 4t_\parallel + 3t_\perp. \tag{16}$$

Info: it is in fact more suggestive to rewrite $h_{1d}^{lattice}(\boldsymbol{k}, k_z)$ as

$$h_{1d}^{lattice}(\boldsymbol{k}, k_z) = \lambda_\perp \sin(k_z c)\tau_x + \left[M_0(\boldsymbol{k}) + 2t_\perp \cos(k_z c)\right]\tau_z,$$

where $M_0(\boldsymbol{k}) = m_0(\boldsymbol{k}) - 2t_\perp$, and correspondingly, the lowest energy solution becomes

$$\begin{aligned} m_1(\boldsymbol{k}) &= \Theta\left[-(2t_\perp + M_0(\boldsymbol{k}))\right](2t_\perp + M_0(\boldsymbol{k})) + \Theta\left[-(2t_\perp - M_0(\boldsymbol{k}))\right](2t_\perp - M_0(\boldsymbol{k})) \\ &= \Theta[|M_0(\boldsymbol{k})| - 2t_\perp](2t_\perp - |M_0(\boldsymbol{k})|), \end{aligned} \tag{17}$$

which shares a symmetric form.

The motivation is attributed as follows: in the first case like in Bi_2Se_3[4], the system owns two Dirac cones at top and bottom surfaces in the film geometry, and when we make it into a semi-magnetic TI with magnetism at only one surface, polarized out of the film plane, the system will be driven into a novel topological phase, exhibiting the half-quantized anomalous Hall effect (HQAHE)[11–13], which is a metallic phase characterized by a half-quantized Hall conductance inside the parity invariant regime near the Γ point, where the single surface Dirac cone emerges. Essentially, the novel topological phase belongs to the quantum anomalous semimetal (QASM)[14] (in the case quantum anomalous metal (QAM) may be more appropriate) with the classifying half-integer index $N_h/2$ where $N_h = 1$. Similar physics is expected to happen with the third case where the only difference is the shift of Dirac cone from Γ to M, and the corresponding semi-magnetic TI will still carry $N_h = 1$ with half-quantized Hall conductance. On the other hand, the realization of $N_h = 2$ will directly lead to an integer quantized Hall conductance, and the corresponding system will lie in metallic quantized anomalous Hall effect (MQAHE) phase[7]. However, in the previous example, the even integer N_h comes from an addition of two separable gapless Dirac cones, each contributing half-quantized Hall conductance as $N_h = 1 + 1$, while a direct inseparable $N_h = 2$ index still lacks of evidence.

Now we propose that a semi-magnetic weak TI (SMWTI) film, for instance $m_0 = 4t_\parallel + 3t_\perp$ without losing generality, shows $N_h = 2$. The origin of this $N_h = 2$ originates from a pair of gapless Dirac cones at X and Y that are connected by their high-energy part, which, however, should be distinguished from the case of graphene[15, 16] where the connected two Dirac cones at K and K' are trivial by themselves without introducing additional mass gaps.

A. Results from Hamiltonian of TI Film

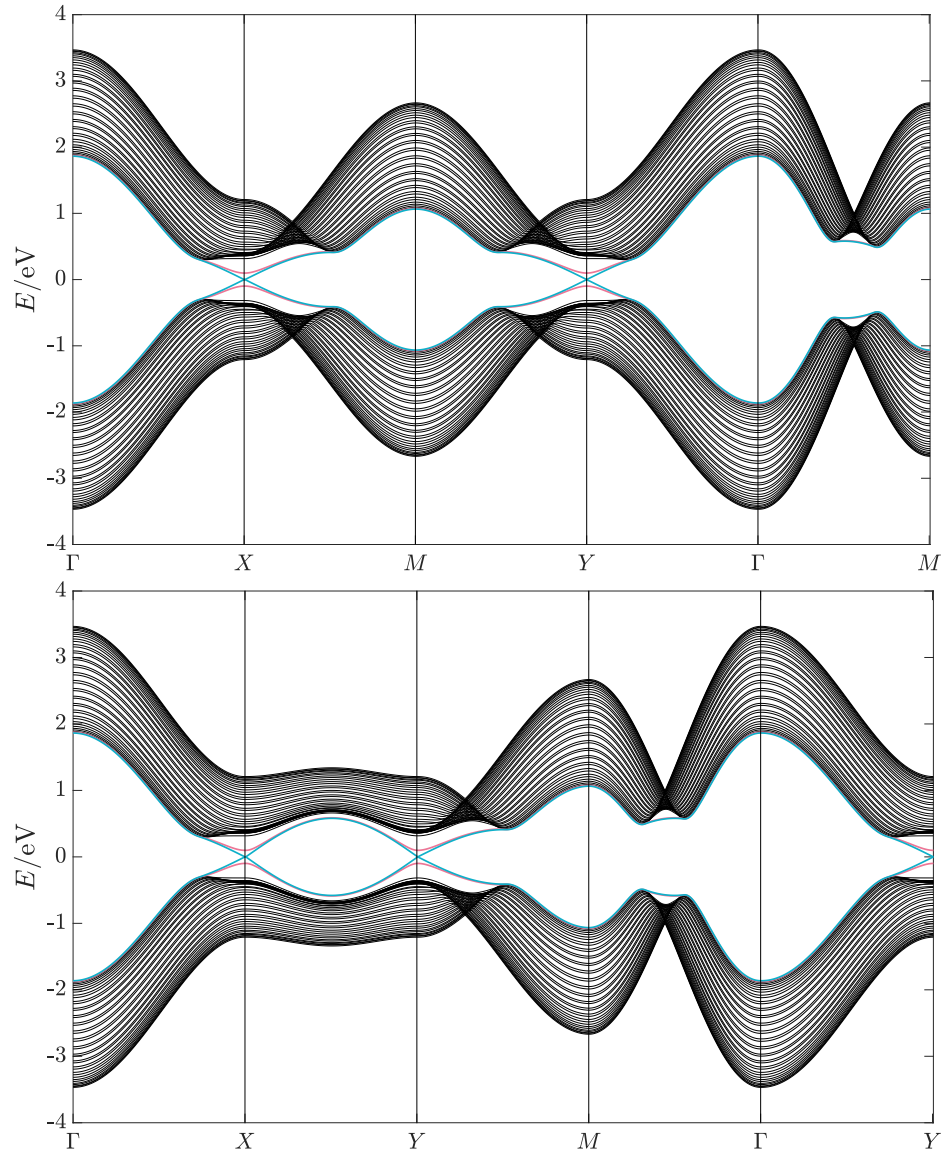


FIG. 1. Band structure of semi-magnetic weak TI film with total layer number $L_z = 20$. Zeeman field is added at top 3 layers with strength $V = 0.1$ eV. Band of gapped and gapless surface states are assigned with red and blue lines, respectively.

The calculated results from thin film tight-binding model are shown in Fig. 1,2,3,4,5.

• Fig. 1 shows band structure of SMWTI along high-symmetry line, where we repeated band structure through X and Y points to show the double-Dirac-cone nature. It is clear that by top-surface magnetism, two cones at the top surface are gapped while two at the

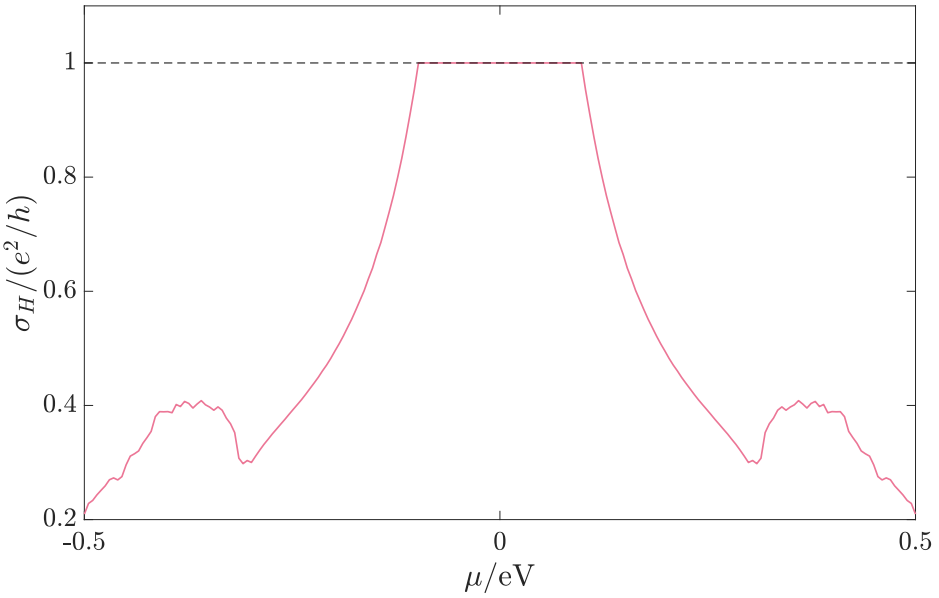


FIG. 2. Hall conductance of semi-magnetic weak TI film with total layer number $L_z = 20$. Zeeman field is added at top 3 layers with strength $V = 0.1$ eV.

bottom remains gapless. Essentially, both gapless (or gapped) Dirac cones are connected by a sharing high-energy part.

• Fig. 2 directly shows quantized Hall conductance as

$$\sigma_H = \frac{e^2}{h}. \tag{18}$$

• Fig. 3 shows Berry curvature distribution for the gapped and gapless band along the high-symmetry line. The Berry curvature for the gapped band shows peak at gapped Dirac points X and Y, but is cancelled by the opposite-sign curvature away from Dirac points, which leads to a cancelled contribution. In contrast, Berry curvature for the gapless band is always positive, which leads to a totally non-vanishing Hall conductance contribution.

• Fig. 4 and Fig. 5 show Berry curvature for gapped and gapless bands in the whole Brillouin zone. The analysis above still works, while a special property needs to be noticed as the large area with $F_{xy}^{\text{gapless}}(k_x, k_y) = 0$ as the Berry curvature for the gapless band. This area not only exists near the two gapless Dirac cones at X and Y but connects them, forming a connected region in Brillouin zone. Especially, such a band structure can be described by a two-band Dirac lattice model as we shall show below, and the zero-curvature area will correspond to the zero-mass area in the two-band model. As shown schematically in Fig. 6, this area geometrically equals to a 2-Torus T^2 with two holes dug out. Since the zero-mass

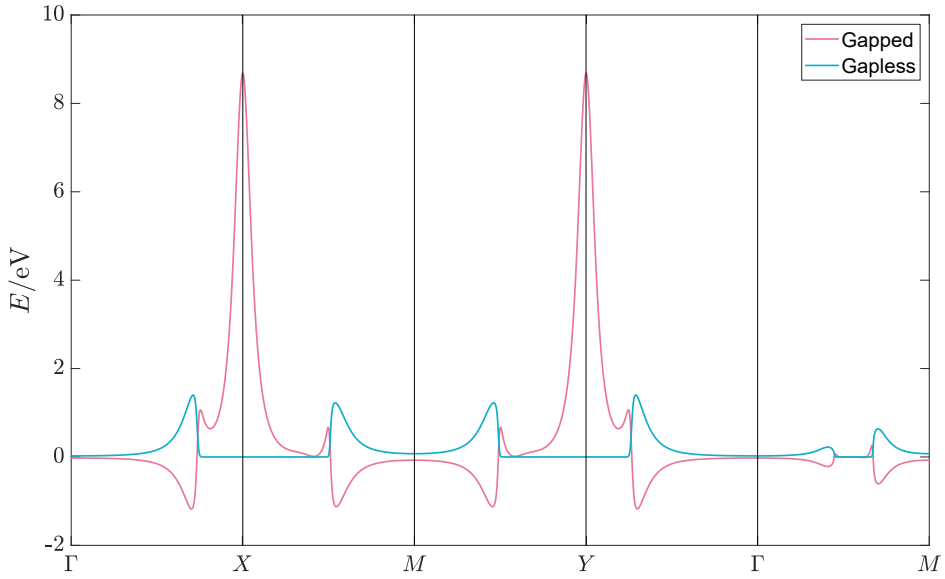


FIG. 3. Berry curvature of semi-magnetic weak TI film with total layer number $L_z = 20$. Zeeman field is added at top 3 layers with strength $V = 0.1$ eV. Curvature of gapped and gapless surface states are assigned with red and blue lines, respectively.

reveals the parity-invariant nature in the area, a claim can be made similar to [13]:

> **For Fermi surface completely confined in the zero-mass/parity-invariant regime, the Hall conductance of this double-gapless Dirac cones band will be e^2/h, so does the whole system.**

Notice that this claim is not a trivial extrapolation, and the proof relies on the topological deformation.

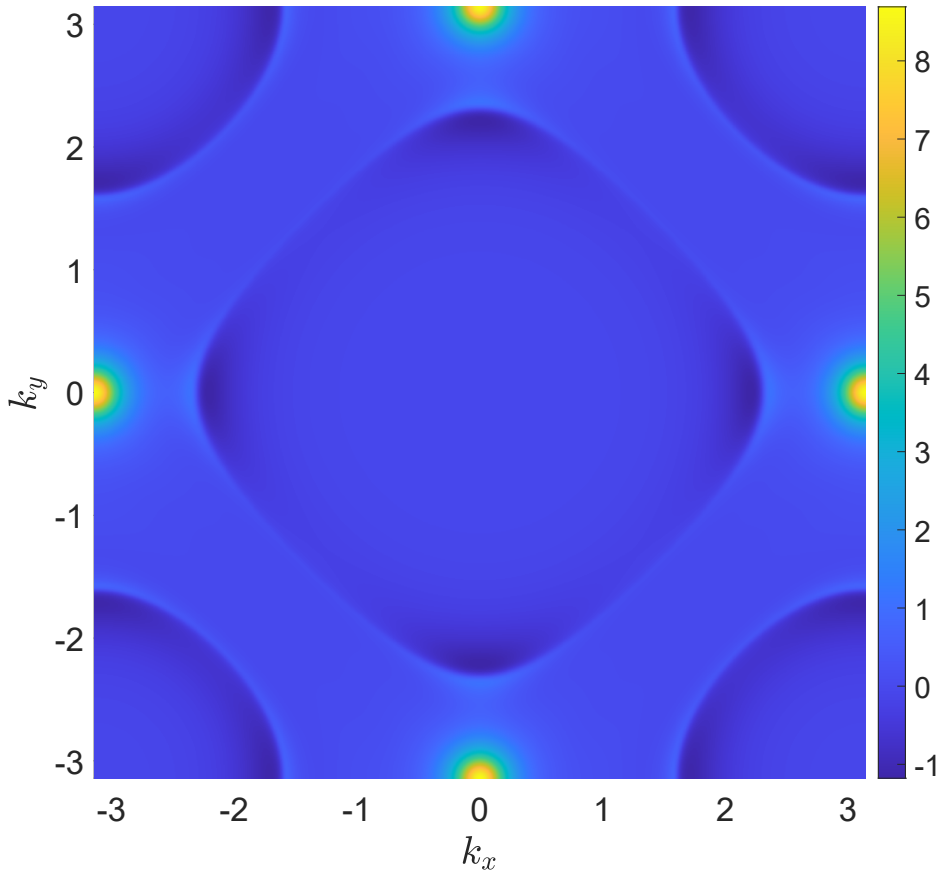


FIG. 4. Berry curvature of semi-magnetic weak TI film with total layer number $L_z = 20$. Zeeman field is added at top 3 layers with strength $V = 0.1$ eV.

B. Results From Effective Model

The proof relies also on the effective two-band model description of the gapless band. For the effective model, by preserving the lowest four bands we still have

$$H_{\rm EFF} = \lambda_{\parallel}(\sin(k_x a)\sigma_x + \sin(k_y a)\sigma_y) + m(\boldsymbol{k})\tau_z\sigma_z + I_S(\boldsymbol{k})\tau_0\sigma_z - I_A(\boldsymbol{k})\tau_y\sigma_z, \tag{19}$$

where the mass $m(\boldsymbol{k})$ is just the lowest eigenvalue of $\mathcal{H}_{1d}$ as

$$m(\boldsymbol{k}) = \Theta\left[-m_0(\boldsymbol{k})\right] m_0(\boldsymbol{k}) + \Theta\left[m_0(\boldsymbol{k}) - 4t_\perp\right](4t_\perp - m_0(\boldsymbol{k})). \tag{20}$$

A more careful look on Eq. (19) tells that we can treat all the latter-three terms as mass terms, since by τ-space diagonalization

$$U_M^\dagger \left[m\tau_z + I_S\tau_0 - I_A\tau_y\right] U_M = \begin{pmatrix} \tilde{m}_+ & \\ & \tilde{m}_- \end{pmatrix}, \tag{21}$$

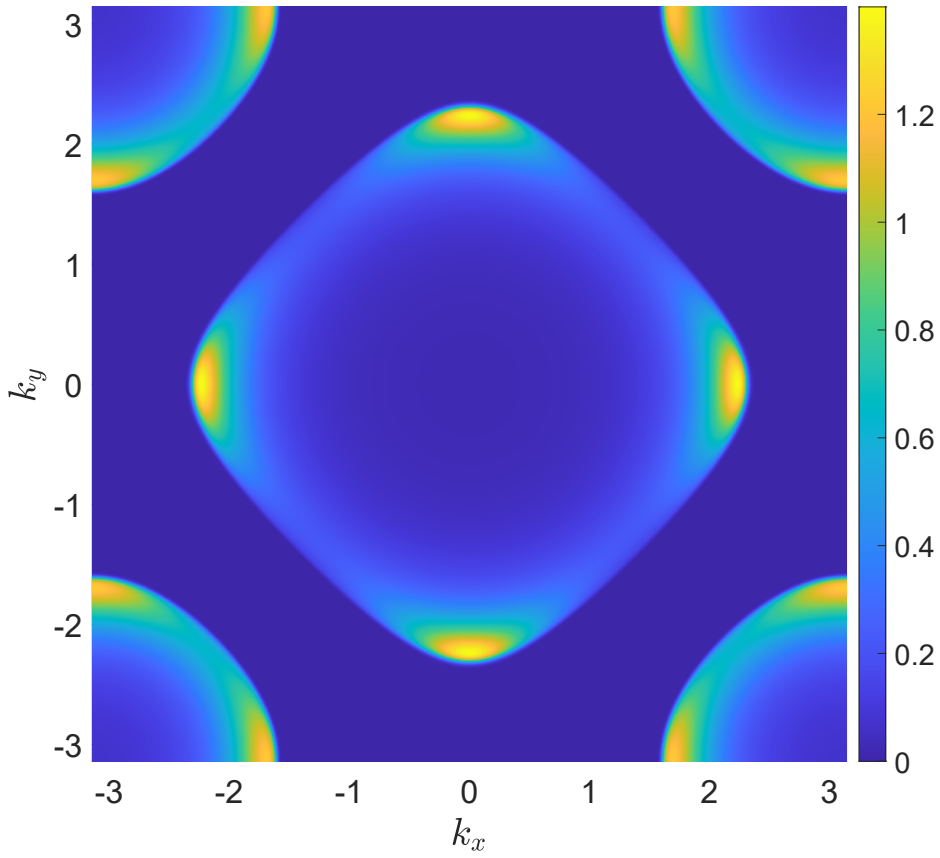


FIG. 5. Berry curvature of semi-magnetic weak TI film with total layer number $L_z = 20$. Zeeman field is added at top 3 layers with strength $V = 0.1$ eV.

where the defined unitary matrix reads

$$U_M = \frac{1}{\sqrt{2}} \begin{pmatrix} i\mathrm{sgn}(I_A)\sqrt{1+\frac{m}{M}} & \sqrt{1-\frac{m}{M}} \\ \sqrt{1-\frac{m}{M}} & i\mathrm{sgn}(I_A)\sqrt{1+\frac{m}{M}} \end{pmatrix}, \tag{22}$$

with $M(\boldsymbol{k}) = \sqrt{m^2(\boldsymbol{k}) + I_A^2(\boldsymbol{k})}$, we can write $\tilde{H}_{\mathrm{EFF}} = \oplus_{\chi=\pm}\tilde{H}_\chi$ with

$$\tilde{H}_\chi = \lambda_\parallel(\sin(k_x a)\sigma_x + \sin(k_y a)\sigma_y) + \tilde{m}_\chi(\boldsymbol{k})\sigma_z, \tag{23}$$

where the effective mass is defined as

$$\tilde{m}_\chi(\boldsymbol{k}) \equiv I_S(\boldsymbol{k}) + \chi\sqrt{m^2(\boldsymbol{k}) + I_A^2(\boldsymbol{k})}. \tag{24}$$

The calculated quantities of effective model are shown in Fig. 7,8,9,10,11,12,13,14,15,16,17,18,19.

• From Fig. 7,13,14, we see that $I_S(\boldsymbol{k})$ and $I_A(\boldsymbol{k})$ are equal at least near surface Dirac cones where their role as low-energy mass-generator is maximal.

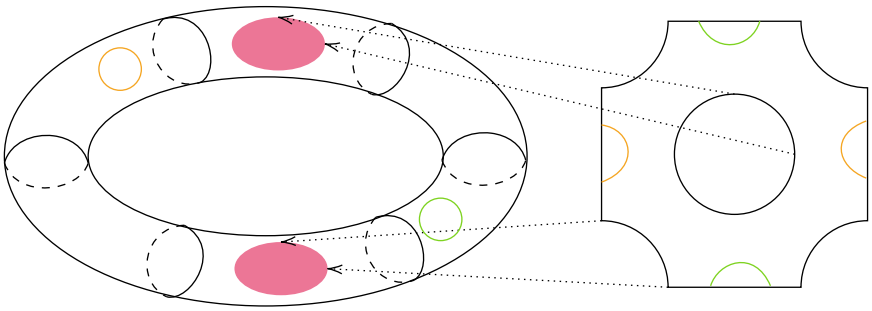


FIG. 6. Schematic picture for zero-mass/curvature area. Red areas on the torus represent two negative mass areas, and mass in other area is zero. Green and orange circles represent the Fermi surface near two gapless Dirac cones.

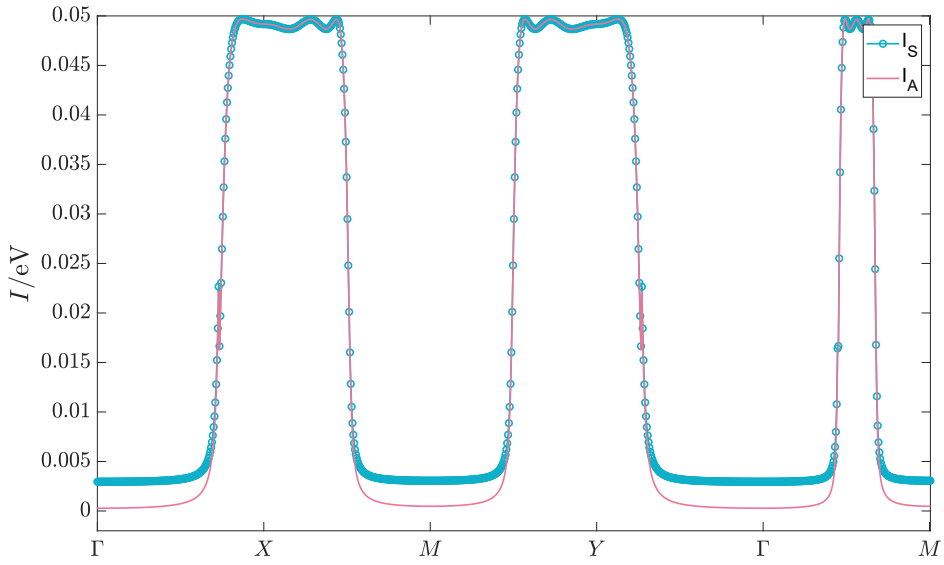


FIG. 7. Symmetric and asymmetric quantities $I_{S/A}(\boldsymbol{k})$ of semi-magnetic weak TI film with total layer number $L_z = 20$. Zeeman field is added at top 3 layers with strength $V = 0.1$ eV.

• The mass configuration Fig. 8,11,12 reveals the existence of two Dirac cones at X and Y.

• The band structure Fig. 9 from effective model corresponds well with the film result.

• The effective mass configuration Fig. 15,16 shows that $\tilde{m}_+(\boldsymbol{k})$ is always positive and is thus trivial, while $\tilde{m}_-(\boldsymbol{k})$ is non-positive with large area where $\tilde{m}_(\boldsymbol{k})$ as effective parity invariant regime, and is thus non-trivial labelled by half-integer topological index.

• The Berry curvature Fig. 17,18,19 shows great correspondence as with the film model.

• Finally, the Hall conductance Fig. 10 directly shows that $\chi = +$ band as the gapped one is trivial, while the gapless $\chi = -$ band is non-trivial with $\sigma_H = N_h e^2/2h$, and $N_h = 2$

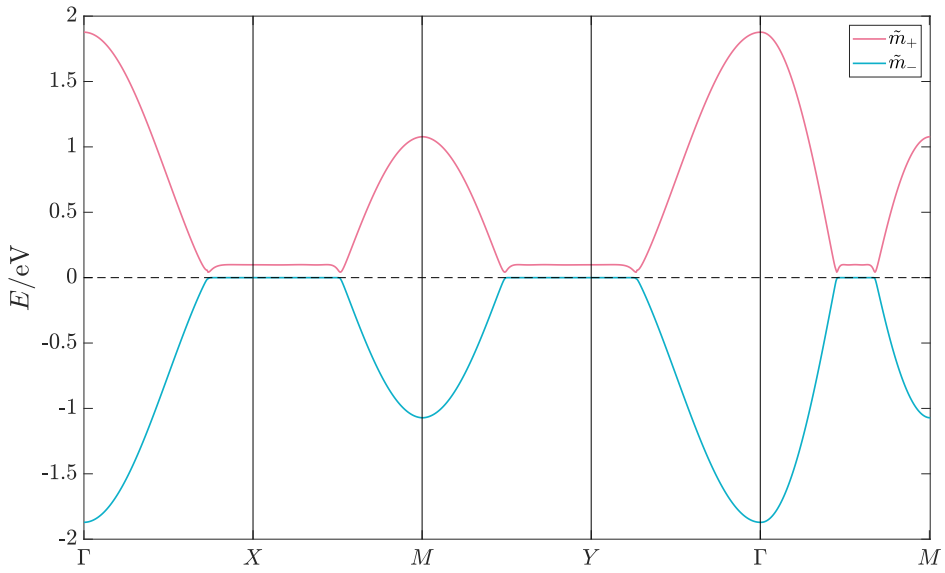


FIG. 8. Lowest effective mass $\tilde{m}_{\pm}(\boldsymbol{k})$ of semi-magnetic weak TI film with total layer number $L_z = 20$. Zeeman field is added at top 3 layers with strength $V = 0.1$ eV.

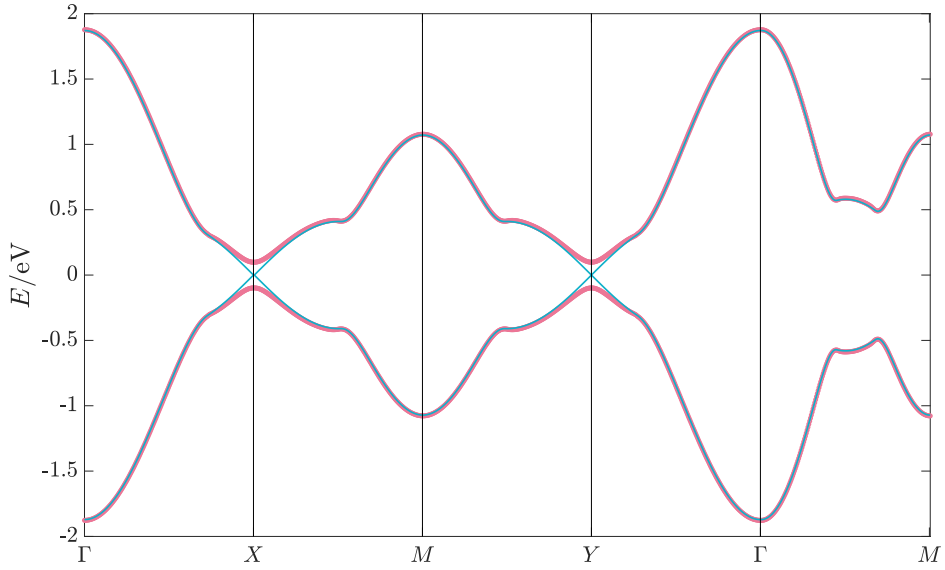


FIG. 9. Band from effective model of semi-magnetic weak TI film with total layer number $L_z = 20$. Zeeman field is added at top 3 layers with strength $V = 0.1$ eV.

which gives rise to an integer quantized Hall conductance.

An explanation from effective low-energy theory by decoupling two Dirac cones can be formed as follows: expand near X and Y points, and one gets four Dirac cones as

$$h_\chi^X(\boldsymbol{q}^X) = -\lambda_\parallel \boldsymbol{q}^X \cdot \bar{\boldsymbol{\sigma}} + \tilde{m}_\chi^X(\boldsymbol{q}^X)\sigma_z, \tag{25a}$$

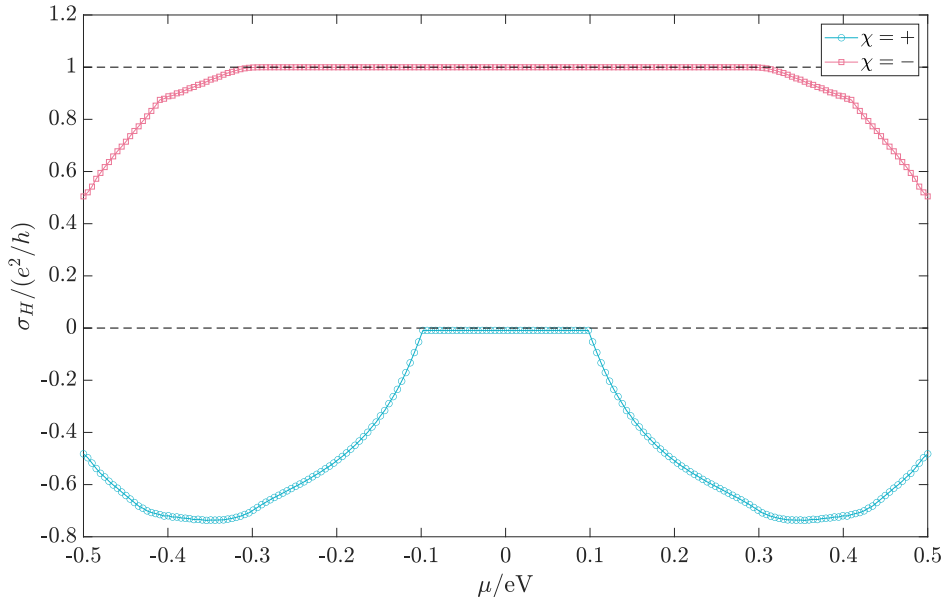


FIG. 10. Hall conductance of $\tilde{H}_\chi$ from effective model of semi-magnetic weak TI film with total layer number $L_z = 20$. Zeeman field is added at top 3 layers with strength $V = 0.1$ eV.

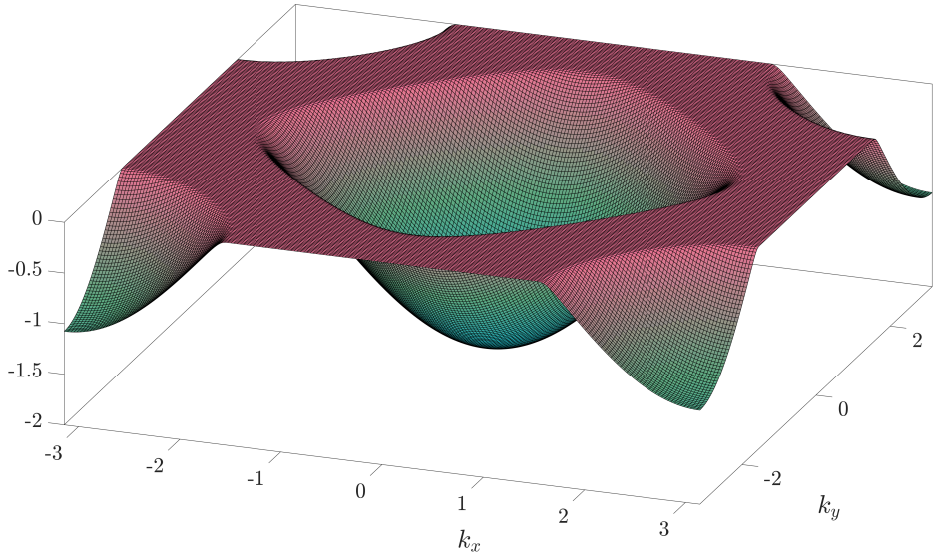


FIG. 11. $m(\boldsymbol{k})$ from solving $\mathcal{H}_{1d}$.

$$h_\chi^Y(\boldsymbol{q}^Y) = \lambda_\parallel \boldsymbol{q}^Y \cdot \bar{\boldsymbol{\sigma}} + \tilde{m}_\chi^Y(\boldsymbol{q}^X)\sigma_z, \tag{25b}$$

where $\boldsymbol{q}^X = \boldsymbol{k} - X = (k_x - \pi, k_y)$, $\boldsymbol{q}^Y = \boldsymbol{k} - Y = (k_x, k_y - \pi)$ describe small momenta near two points, and we have defined $\bar{\boldsymbol{\sigma}} = (\sigma_x, -\sigma_y)$. Masses for the four cones are defined as

$$\tilde{m}_+^{X/Y}(\boldsymbol{q}^{X/Y}) = V + \sqrt{V^2 + (\tilde{m}_-^{X/Y}(\boldsymbol{q})^{X/Y})^2}, \tag{26a}$$

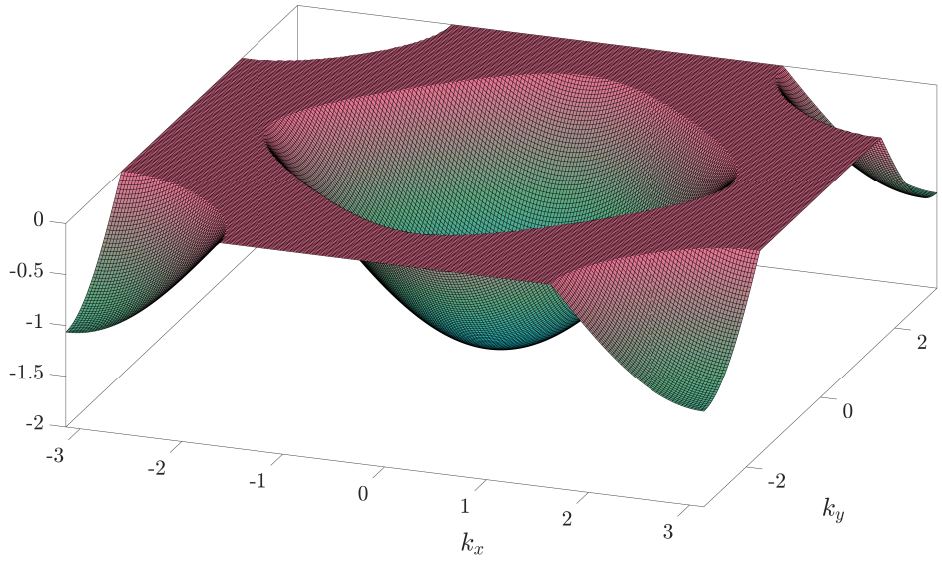


FIG. 12. $m(\boldsymbol{k})$ from analytic expression Eq. (20).

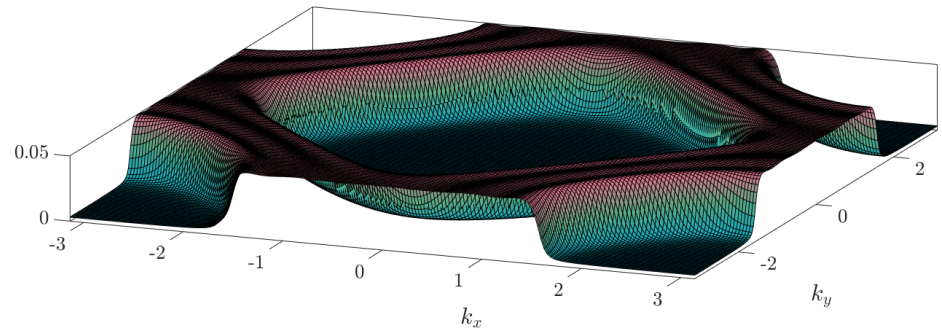


FIG. 13. $I_S(\boldsymbol{k})$.

$$\tilde{m}_{-}^{X/Y}(\boldsymbol{q}^{X/Y}) = \Theta(4t_{\parallel} - m_0 + t_{\parallel}(q^{X/Y})^2)\left[(m_0 - 4t_{\parallel}) - t_{\parallel}(q^{X/Y})^2\right], \tag{26b}$$

where $V = I_S = I_A \approx V_z^{\text{top}}/2 > 0$ is assumed. Expressions, combined with the Hall conductance expression[13, 17] in the case (X/Y ignored)

$$\begin{aligned}\sigma_H^{\chi} &= \frac{e^2}{2h}\left[\text{sgn}(\tilde{m}_{\chi}(q^{\chi} \to +\infty)) - \cos\left(\theta_{q_F^{\chi}}\right)\right] \\ &= \frac{e^2}{2h}\left[\chi - \frac{\tilde{m}_{\chi}(q_F^{\chi})}{\sqrt{\lambda_{\parallel}^2(q_F^{\chi})^2 + (\tilde{m}_{\chi}(q_F^{\chi}))^2}}\right],\end{aligned} \tag{27}$$

lead to conclusion that gapped $\chi = +$ cones for X/Y with $q_F^{\chi} = 0$ are trivial for Fermi surface lies inside the Zeeman gap, while $\chi = -$ cones as two gapless one are non-trivial

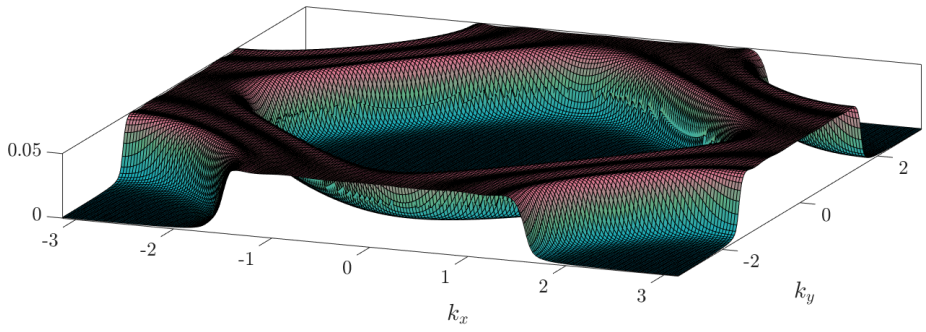


FIG. 14. $I_A(\boldsymbol{k})$.

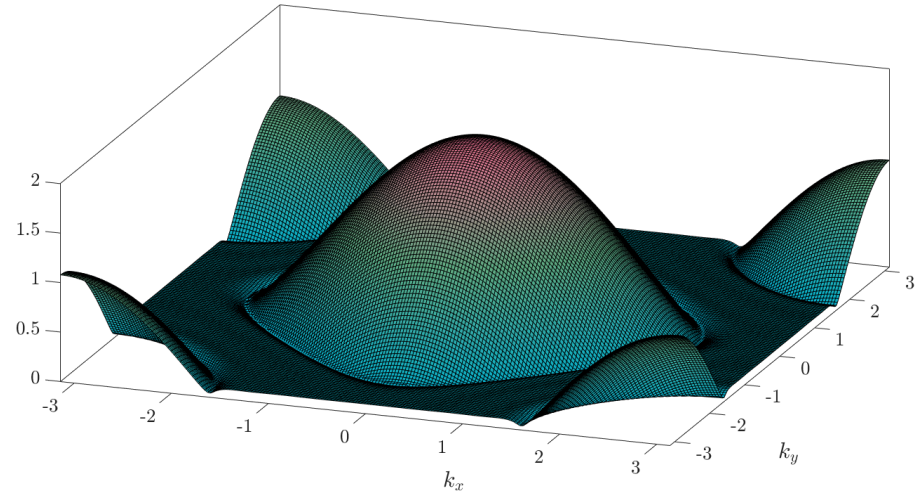


FIG. 15. $\tilde{m}_+(\boldsymbol{k})$, the low-mass area does not touch zero.

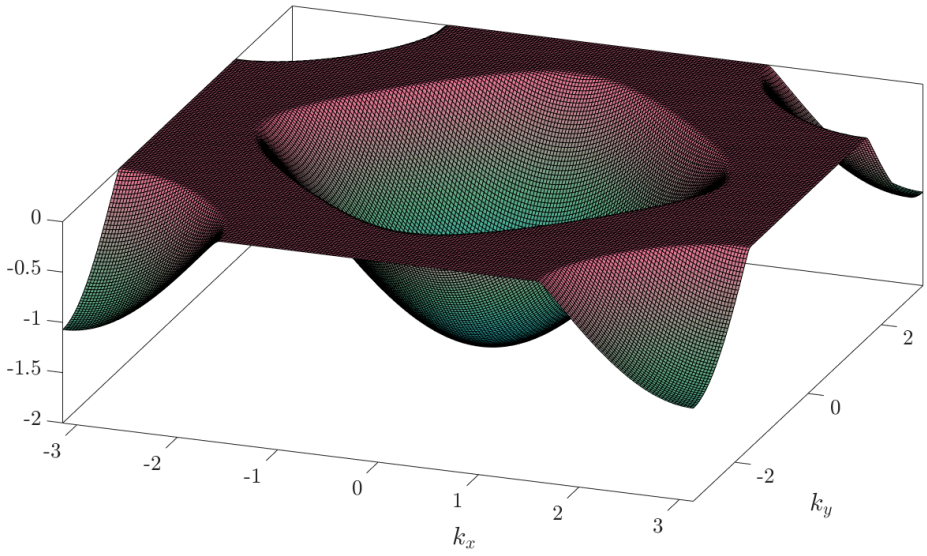


FIG. 16. $\tilde{m}_-(\boldsymbol{k})$.

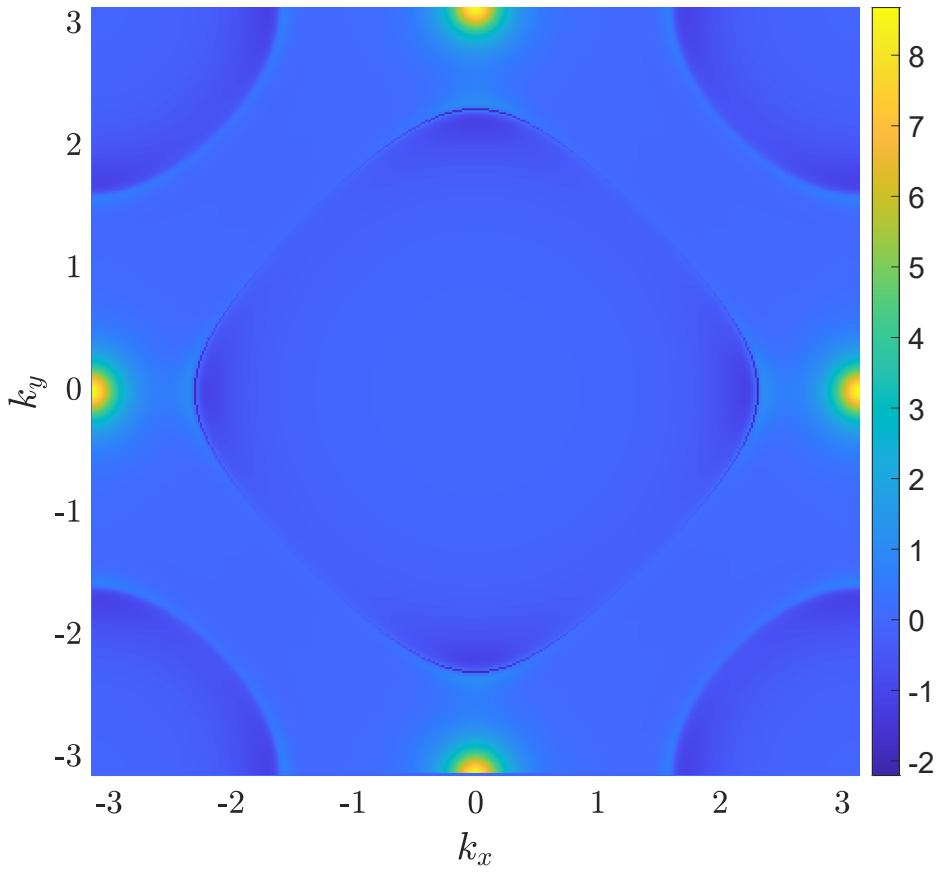


FIG. 17. Berry curvature of gapped surface band from effective model.

with both contributing $e^2/2h$ to the Hall conductance when

$$\tilde{m}_{-}^{X/Y}(q_F^{-,X/Y}) = 0 \tag{28}$$

is satisfied as the Fermi surface lies inside parity invariant regime of two gapless Dirac cones.

The four-cone picture above is suggestive, but it breaks the connectivity of the paired gapped/gapless Dirac cones. Now we prove the exact quantization based on topological deformation of Fermi surface, as illustrated in Fig. 20. The rationality behind roots in the vanishing of Berry curvature, which originates from the vanishing of mass term in the two-band gapless-Dirac-cone model. Specifically speaking, for a reduced lattice Dirac two-band model

$$h = \boldsymbol{d} \cdot \boldsymbol{\sigma} = \sin(k_x)\sigma_x + \sin(k_y)\sigma_y + m(\boldsymbol{k})\sigma_z, \tag{29}$$

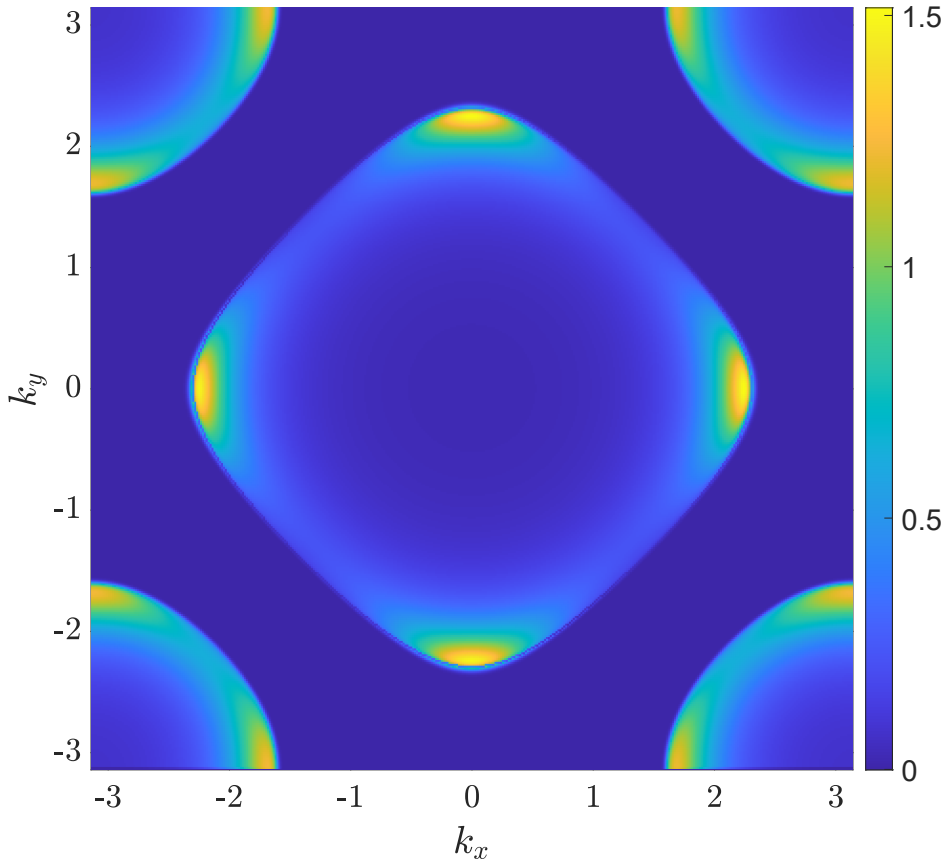


FIG. 18. Berry curvature of gapless surface band from effective model.

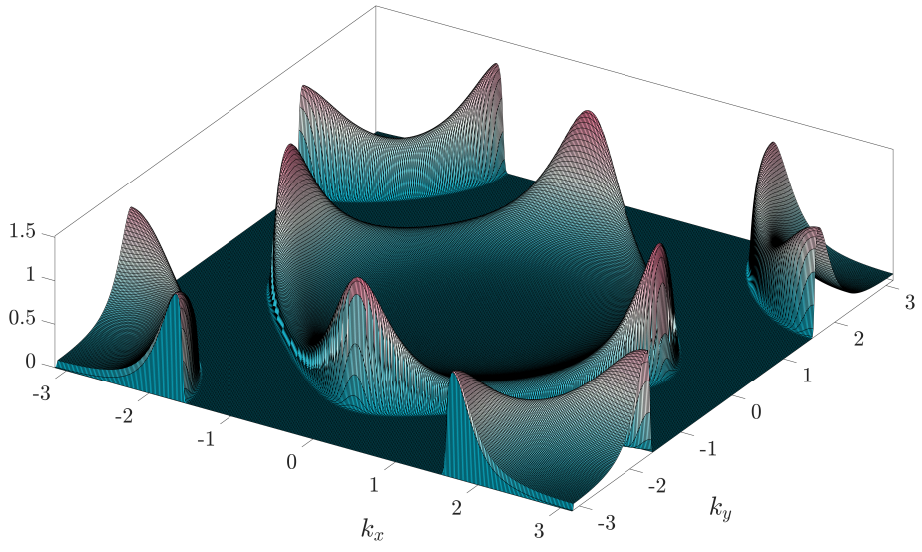


FIG. 19. 3-D Berry curvature of gapless surface band from effective model.

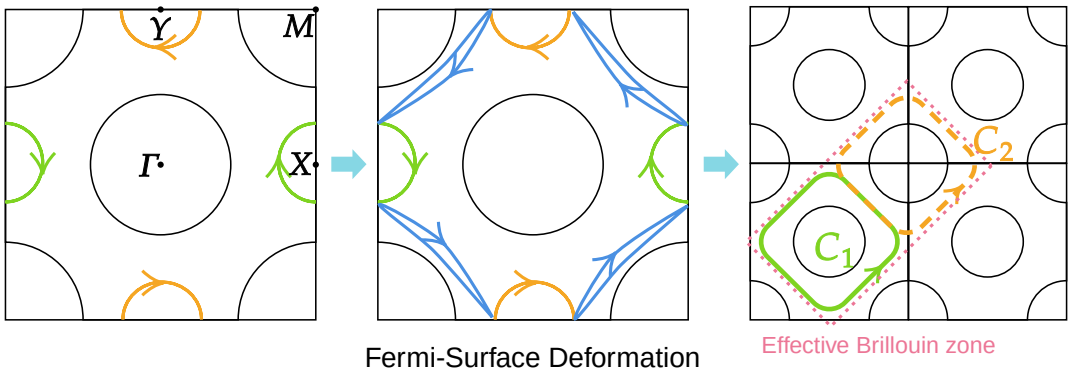


FIG. 20. Deformation of Fermi surface. This deformation turns Berry phase line integral around Fermi surface, as green and origin circles around gapless Dirac points X and Y, equivalently into two circle integrals on $C_{1,2}$ around the areas with non-vanishing mass/curvature.

its Berry curvature of the valence band equals

$$\begin{aligned} F_{xy}(\boldsymbol{k}) &= \frac{1}{2}(\partial_{k_x}\hat{\boldsymbol{d}} \times \partial_{k_y}\hat{\boldsymbol{d}}) \cdot \hat{\boldsymbol{d}} = \frac{1}{2}\frac{(\partial_{k_x}\boldsymbol{d} \times \partial_{k_y}\boldsymbol{d}) \cdot \boldsymbol{d}}{d^3} \\ &= \frac{1}{2}\frac{[\cos(k_x)\cos(k_y) - \sin(k_x)\cos(k_y)\partial_{k_x} - \sin(k_y)\cos(k_x)\partial_{k_y}]m(\boldsymbol{k})}{(\sin^2(k_x) + \sin^2(k_y) + m^2(\boldsymbol{k}))}, \end{aligned} \tag{30}$$

then immediately one sees that the contribution to the curvature is attributed to the non-vanishing m and its first-order derivative, and the area mass $m = 0$ directly leads to the zero-curvature area (ZCA), i.e., the parity invariant regime (PIR). Formally,

$$\text{ZCA} = \{\boldsymbol{k} | F_{xy}(\boldsymbol{k}) = 0\}. \tag{31}$$

Notice that ZCA is an open set is the Brillouin zone (BZ) since zero-mass boundary can contribute non-zero derivative, then we conclude that

$$\text{ZCA} = \text{PIR} - \partial\text{PIR} - \{X, Y\}, \tag{32}$$

where a single ∂ acts on an area and gets its boundary. Notice that we also throw set of Dirac points where the curvature is singular.

Now to explain the quantization, let us assume that the chemical potential μ lies inside the Zeeman gap so that the trivial gapped Dirac cones do not contribute, and the Fermi surface (FS) defined by

$$\text{FS} = \{\boldsymbol{k} | \mu = d(\boldsymbol{k})\} \tag{33}$$

completely lies inside the zero-curvature zone. Then for zero-temperature Hall conductance of the gapless Dirac cone (we have assumed that the Fermi surface crosses with the valence band, without losing generality)

$$\sigma_H = -\frac{e^2}{2\pi h}\int_{\mathrm{BZ}} \mathrm{d}^2k\Theta(\mu - d(\boldsymbol{k}))F_{xy}(\boldsymbol{k}) = -\frac{e^2}{2\pi h}\int_{d(\boldsymbol{k})<\mu} \mathrm{d}^2kF_{xy}(\boldsymbol{k}), \tag{34}$$

it will be legal to subtract arbitrary area inside the zero-curvature area $\Delta A \subseteq \mathrm{ZCA}$ from the original integral area $A_O = \{\boldsymbol{k}|d(\boldsymbol{k}) < \mu\}$, whose boundary is the Fermi surface $\partial A_O = \mathrm{FS}$, i.e., by denoting the subtracted area as $\tilde{A} = A_O - \Delta A$, we have

$$\sigma_H = -\frac{e^2}{2\pi h}\int_{A_O} \mathrm{d}^2kF_{xy}(\boldsymbol{k}) = -\frac{e^2}{2\pi h}\int_{\tilde{A}} \mathrm{d}^2kF_{xy}(\boldsymbol{k}). \tag{35}$$

Essentially, we can choose ΔA to be four areas in the standard Brillouin zone whose boundaries connect the non-path-connected Fermi surface located around X and Y, as shown in the second picture of Fig. 20, where the orientation is chosen such that the area to be integrated lies at the left of the boundary traveling along the given orientation, as represented by the arrow. And the integral is now bounded by $\partial\tilde{A} = \mathrm{FS} - \partial(\Delta A)$, as represented by green, orange and blue circles (by periodicity of Brillouin zone). Essentially, we notice that by splitting the three circles at their crossing points, and re-combining them by the correct orientation, we get two circles C_1 and C_2 circling around non-parity invariant area near Γ and M points, respectively, as shown in the third picture. Mathematically, by the stokes theorem, this is equivalent to say

$$\sigma_H = -\frac{e^2}{2\pi h}\oint_{\mathrm{FS}} \mathrm{d}\boldsymbol{k}\cdot\boldsymbol{A}(\boldsymbol{k}) = -\frac{e^2}{2\pi h}\oint_{\mathrm{FS}-\partial(\Delta A)} \mathrm{d}\boldsymbol{k}\cdot\boldsymbol{A}(\boldsymbol{k}) = -\frac{e^2}{2\pi h}\oint_{C_1+C_2} \mathrm{d}\boldsymbol{k}\cdot\boldsymbol{A}(\boldsymbol{k}), \tag{36}$$

with Berry connection $\nabla_k \times \boldsymbol{A} = F_{xy}$ identified. Now, the Hall conductance is split into two parts as

$$\sigma_H = \sigma_H^1 + \sigma_H^2, \tag{37a}$$

$$\sigma_H^l = -\frac{e^2}{2\pi h}\oint_{C_l} \mathrm{d}\boldsymbol{k}\cdot\boldsymbol{A}^l(\boldsymbol{k}),\ l = 1, 2, \tag{37b}$$

with $C_l \subseteq \mathrm{ZCA}$ circles lying inside the zero-curvature area assigned with unit winding number around different high-symmetry points

$$\omega_1(\Gamma) = \omega_2(M) = 1. \tag{38}$$

Notice that the above two requirements completely define the C-circles, without considering their detailed analytic expression in Brillouin zone, i.e., they belong to the homotopy class which is the $(1,1,0,0)$ element of the Abelian group composed of independent windings around of the Γ, M, X, Y points (the winding lines need to lie inside ZCA)

$$\bigoplus_{x\in\{\Gamma,M,X,Y\}} \pi_1(R(x)) = \mathbb{Z}\oplus\mathbb{Z}\oplus\mathbb{Z}\oplus\mathbb{Z}, \tag{39}$$

with $R(x)$ simply referring to loop with only high-symmetry point x inside, and $R(x)$ lies inside the zero-curvature area. Notice that it is not the non-Abelian first homotopy group of the zero-curvature area, a two-torus with four holes at high-symmetry points. The reason behind the identification for the homotopy class is again attributed to the arbitrarily extendable nature of the integral area in the zero-curvature area, or equivalently, triviality of trivial-loop Berry phase integral inside this area,

$$\int_{\Delta A\subseteq \mathrm{ZCA}} \mathrm{d}^2k F_{xy} = \oint_{\partial(\Delta A)=C\subseteq \mathrm{ZCA}} \mathrm{d}\boldsymbol{k}\cdot\boldsymbol{A} = 0, \tag{40}$$

where the trivial-loop $C=\partial(\Delta A)$ is the boundary of a **simply-connected area** $\Delta A\subseteq$ ZCA. An interesting point to notice is that our deformation process changes the homotopy class of the integral boundary, from the Fermi surface to $C_{1,2}$, belonging to $(0,0,1,1)$ and $(1,1,0,0)$ classes, respectively. Essentially notice that C_2 can not be continuously deformed into $\bar{C}_1$, obstructed by the existing singularities X and Y.

To see the quantized nature, notice that each Berry connection loop integral yields a half-quantized value[13, 14] when the integral loop lies inside the parity-invariant regime, then we can attach

$$\frac{1}{2\pi}\oint_{C_1}\mathrm{d}\boldsymbol{k}\cdot\boldsymbol{A}^1(\boldsymbol{k}) = \frac{N_1}{2}, \tag{41a}$$

$$\frac{1}{2\pi}\oint_{C_2}\mathrm{d}\boldsymbol{k}\cdot\boldsymbol{A}^2(\boldsymbol{k}) = \frac{N_2}{2}, \tag{41b}$$

and the Hall conductance becomes

$$\sigma_H = -\frac{e^2}{2\pi h}\oint_{C_1+C_2}\mathrm{d}\boldsymbol{k}\cdot\boldsymbol{A}(\boldsymbol{k}) = -\frac{e^2}{h}\frac{N_1+N_2}{2}, \tag{42}$$

which is to be another half-integer. It is reasonable to assign $N_1=N_2=-1$ in our case.

The Berry phase over $C_{1,2}$ can be carried out explicitly. Given Hamiltonian $H=\boldsymbol{h}\cdot\boldsymbol{\sigma}$, its lower band wavefunction is expressed as

$$u_- = \frac{1}{\sqrt{2}}\begin{pmatrix} -e^{-i\phi}\sqrt{1+h_3/h} \\ \sqrt{1-h_3/h}\end{pmatrix}, \tag{43}$$

with $h = |\boldsymbol{h}|$, $\tan\phi = h_y/h_x$. In our two-band model describing the gapless Dirac band,

$$\boldsymbol{h} = (\lambda_\parallel \sin(k_x), \lambda_\parallel \sin(k_y), m(\boldsymbol{k})), \tag{44}$$

and possible not removable singularity for ϕ occurs at four high-symmetry points (Γ, X, Y, M), with the limit-taking of ϕ varies with approaching direction and makes it undefined. Nevertheless, notice that C_1 and C_2 are loops surrounding points Γ and M in their positive orientations, respectively, we then only need to make sure that the expression Eq. (43) is singularity-free in areas around Γ and M. This is indeed the case since we have verified that $h_3 = m(\boldsymbol{k}) \leq 0$ in the whole Brillouin zone, and especially $h_3 < 0$ at Γ and M, then since now

$$h(\Gamma/M) = |h_3(\Gamma/M)| = -h_3(\Gamma/M), \tag{45}$$

we find that Eq. (43) reads

$$u_-(\Gamma/M) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{46}$$

which is well-defined, and the singularities of u_- at Γ and M become removable. Now inside the zero-curvature area with $h_3 = m(\boldsymbol{k}) \equiv 0$, the Berry connection of u_- reads

$$A_\mu = i \langle u_- | \partial_{k_\mu} | u_- \rangle = \frac{1}{2} \partial_{k_\mu} \phi, \tag{47}$$

and the Berry phase over loop becomes

$$\oint_{C_l} \mathrm{d}\boldsymbol{k} \cdot \boldsymbol{A}^l = \frac{1}{2} \sum_s \left[\phi_l(k_s^+) - \phi_l(k_s^-) \right], \tag{48}$$

where $l = 1, 2$ represents different loops, k_s refers to singularities of the phase ϕ_l met on loop C_l, and $k_s^\pm$ are infinitesimal for-/backward values over k_s on the loop. The $\pm$-limit are required to be well-defined separately.

Now for C_1, its ϕ-singularities appear at two points

$$k_{s1} = (0, \alpha) \tag{49a}$$

$$k_{s2} = (0, -\alpha), \tag{49b}$$

with $\alpha > 0$ and makes k_s lie inside the zero-curvature area, and from Fig. 5 possibly $\alpha \sim 2.5$. The for-/backward infinitesimal values referred to singularities above on C_1 can be defined as

$$k_{s1}^\pm = (\mp\epsilon, \alpha - \epsilon), \tag{50a}$$

$$k_{s2}^{\pm} = (\pm\epsilon, -\alpha + \epsilon), \tag{50b}$$

with $\epsilon \to 0+$ respecting the loop orientation. Correspondingly,

$$\phi(k_{s1}^{\pm}) = \arctan \frac{\sin(\alpha - \epsilon)}{\sin(\mp\epsilon)} = \mp\frac{\pi}{2}, \tag{51a}$$

$$\phi(k_{s2}^{\pm}) = \arctan \frac{\sin(-\alpha + \epsilon)}{\sin(\pm\epsilon)} = \mp\frac{\pi}{2}, \tag{51b}$$

which leads to

$$\oint_{C_1} \mathrm{d}\boldsymbol{k} \cdot \boldsymbol{A}^1 = \frac{1}{2}\left[\left(-\frac{\pi}{2} - \frac{\pi}{2}\right) + \left(-\frac{\pi}{2} - \frac{\pi}{2}\right)\right] = -\pi, \tag{52}$$

and gives $N_1 = -1$. Similarly, for C_2, singularities appear at

$$k_{s1} = (\pi, \alpha) \tag{53a}$$

$$k_{s2} = (\pi, -\alpha), \tag{53b}$$

with $\alpha \sim 1$, and

$$k_{s1}^{\pm} = (\mp(\pi - \epsilon), \alpha + \epsilon),\ \phi(k_{s1}^{\pm}) = \mp\frac{\pi}{2}, \tag{54a}$$

$$k_{s2}^{\pm} = (\pm(\pi - \epsilon), -\alpha - \epsilon),\ \phi(k_{s2}^{\pm}) = \mp\frac{\pi}{2}, \tag{54b}$$

which leads to

$$\oint_{C_2} \mathrm{d}\boldsymbol{k} \cdot \boldsymbol{A}^2 = -\pi,\ N_2 = -1. \tag{55}$$

Totally, the Hall conductance reads

$$\sigma_H = -\frac{e^2}{2\pi h} \oint_{C_1 + C_2} \mathrm{d}\boldsymbol{k} \cdot \boldsymbol{A}(\boldsymbol{k}) = -\frac{e^2}{h}\frac{N_1 + N_2}{2} = \frac{e^2}{h}. \tag{56}$$

Notice that choosing $C_{1,2}$ as Berry phase loops has advantage than original Fermi surface circles, since that latter requires us to assign more about the undefined singularities, say Y-singularity for FS-loop around X, and vice versa. However, by treating two FS-loop-integral together, we can still assign well-defined wavefunction inside integral area by orientation, and the same result will be got.

C. Lifshitz Transitions with Varying m_0

With z-open boundary condition that makes the 3-D TI a film, which is effectively 2-D, the topology of the Fermi surface in this system will change when varying m_0 as Lifshitz transitions, which gives rise to topological phase transitions between strong, weak and trivial topological phases with change of Dirac points. The change of the Fermi surface can indeed be reflected by the change of lowest mass

$$m_1(\boldsymbol{k}) = \Theta\left[-m_0(\boldsymbol{k})\right] m_0(\boldsymbol{k}) + \Theta\left[m_0(\boldsymbol{k}) - 4t_\perp\right](4t_\perp - m_0(\boldsymbol{k})), \tag{57}$$

when we fix the chemical potential μ.

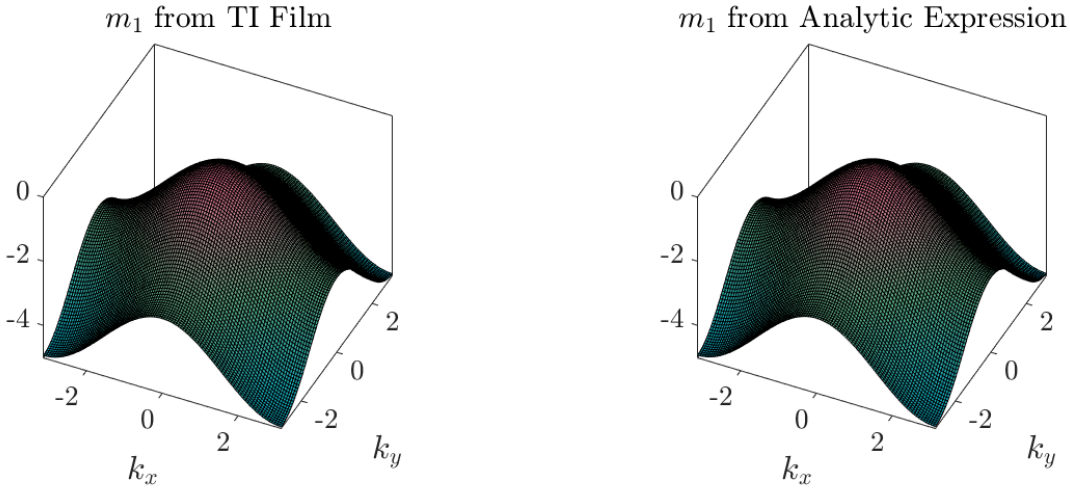


FIG. 21. Mass of the lowest Dirac cone in a TI film at $m_0 = -t_\perp$. A trivial insulator in the z-open 2-D case.

As shown in Fig. 21,22,23,24,25,26,27, the static mass m_0 of the TI film and lowest Dirac

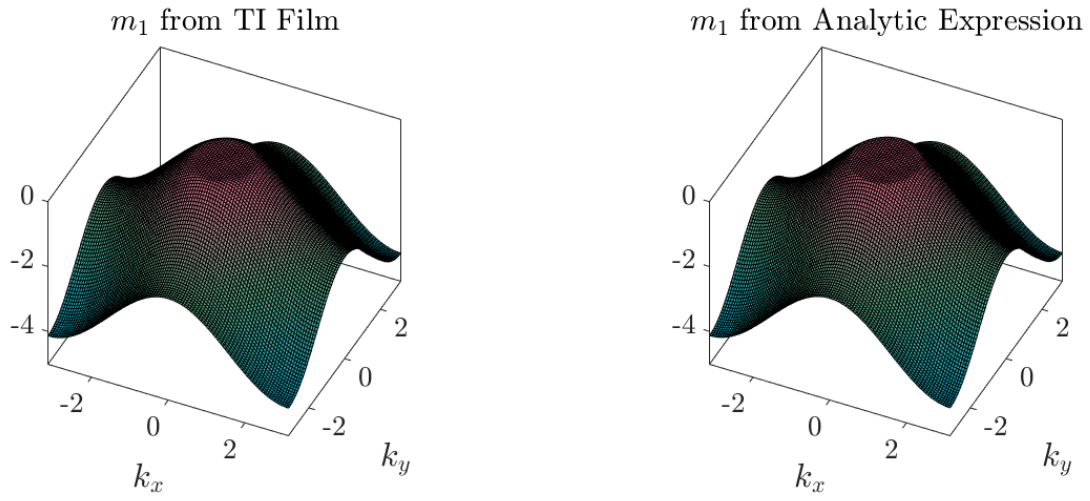


FIG. 22. Mass of the lowest Dirac cone in a TI film at $m_0 = t_\perp$. Single Dirac cone at Γ.

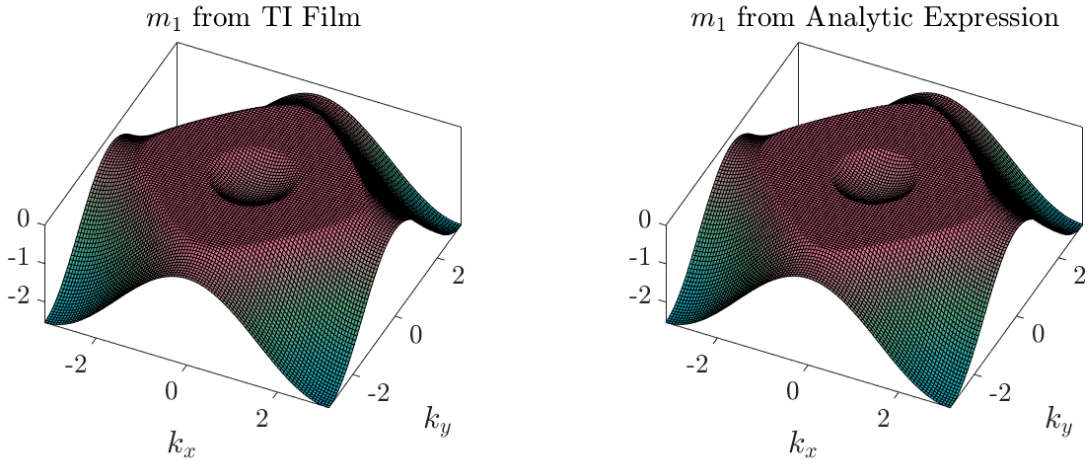


FIG. 23. Mass of the lowest Dirac cone in a TI film at $m_0 = 5t_\perp$. A trivial insulator in the z-open 2-D case.

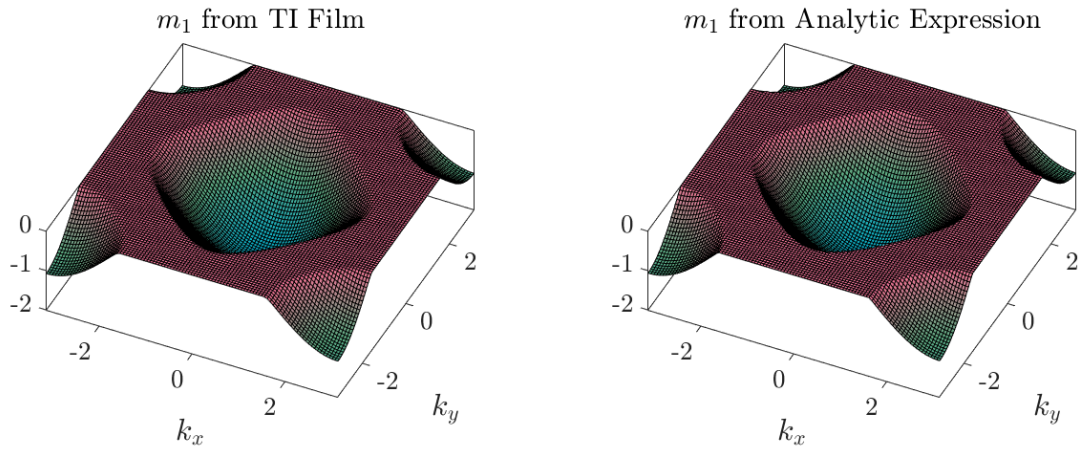


FIG. 24. Mass of the lowest Dirac cone in a TI film at $m_0 = 4t_{\parallel} + 3t_{\perp}$. Connected pair Dirac cones at X and Y.

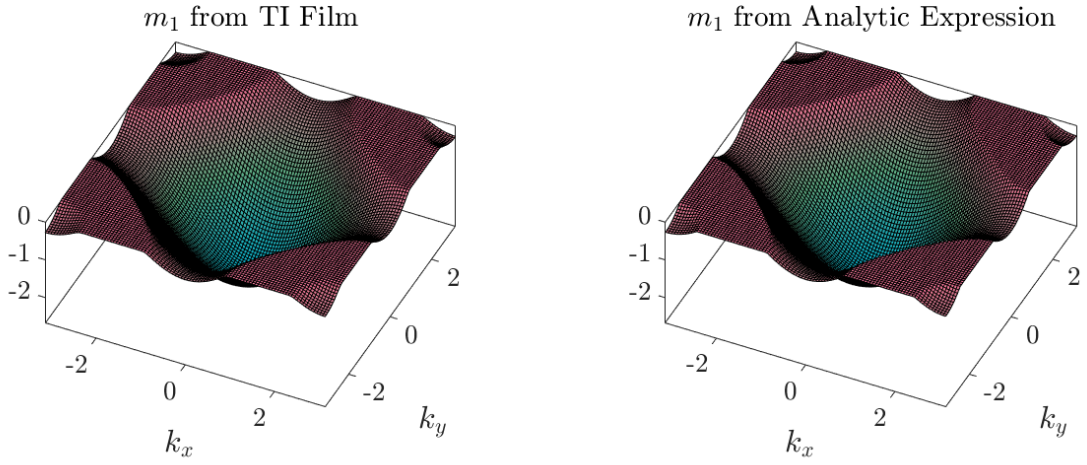


FIG. 25. Mass of the lowest Dirac cone in a TI film at $m_0 = 4t_{\parallel} + 5t_{\perp}$. A trivial insulator in the z-open 2-D case.

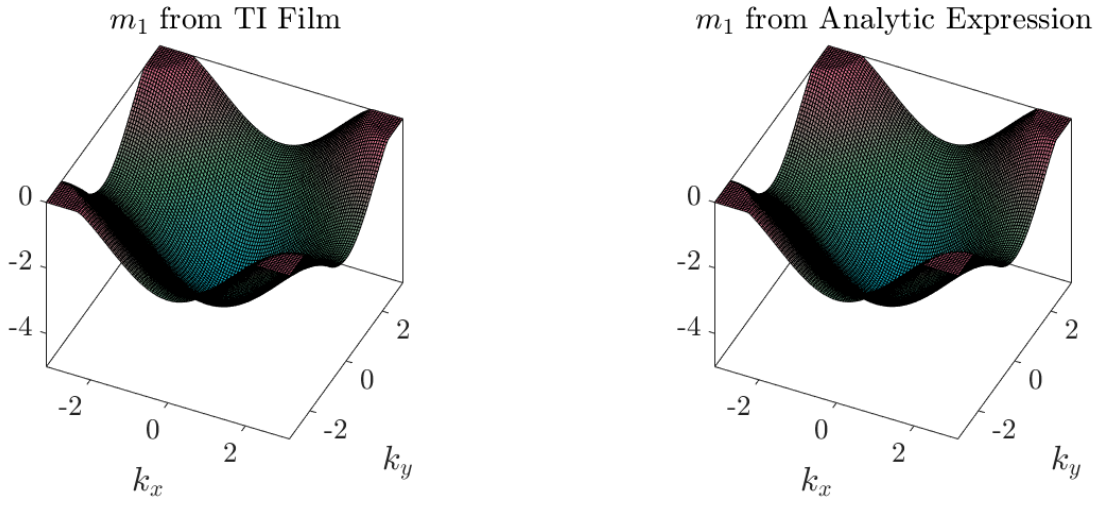


FIG. 26. Mass of the lowest Dirac cone in a TI film at $m_0 = 8t_{\parallel} + 3t_{\perp}$. Single Dirac cone at M.

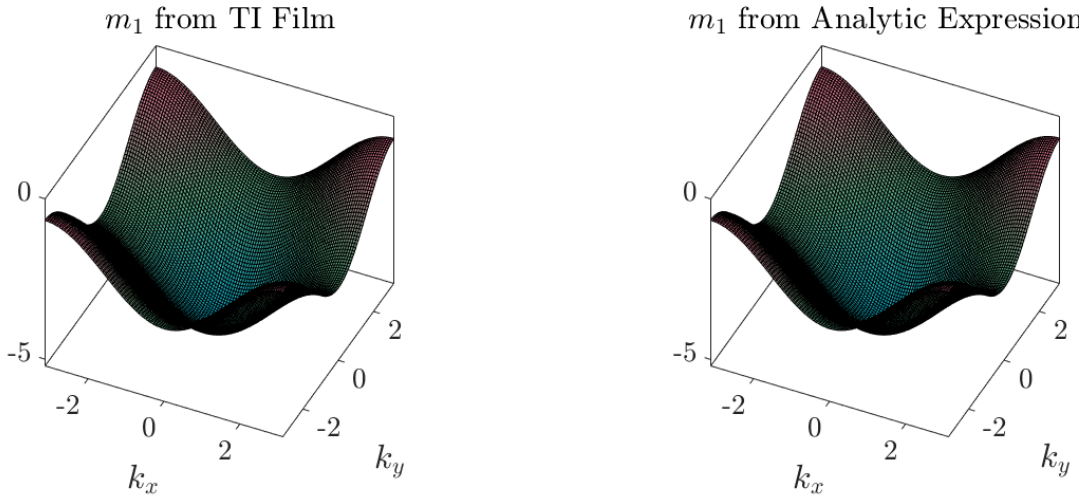


FIG. 27. Mass of the lowest Dirac cone in a TI film at $m_0 = 12t_{\parallel}$. A trivial insulator in the z-open 2-D case.

mass $m_1(\boldsymbol{k})$ have correspondence as

$$m_0 \in \begin{cases} (-\infty, 0) \\ (0, 4t_\perp) \\ (4t_\perp, 4t_\parallel) \\ (4t_\parallel, 4t_\parallel + 4t_\perp) \\ (4t_\parallel + 4t_\perp, 8t_\parallel) \\ (8t_\parallel, 8t_\parallel + 4t_\perp) \\ (8t_\parallel + 4t_\perp, +\infty) \end{cases}, \text{ shape of PIR with } m_1(\boldsymbol{k}) = 0 \begin{cases} \text{non} \\ \text{disk at } \Gamma \\ \text{ring around } \Gamma \\ \text{connected disks at} X, Y \\ \text{ring around } M \\ \text{dist at } M \\ \text{non} \end{cases} . \tag{58}$$

For the case in the middle, we have deliberately used terms at X and Y to emphasis the Dirac cones at these two points, while a more appropriate description is 'Torus with two holes at Γ and M'.

For the Lifshitz transition, say now we put $\mu \to 0_-$, then the Fermi surface will be

$$m_0 \in \begin{cases} (-\infty, 0) \\ (0, 4t_\perp) \\ (4t_\perp, 4t_\parallel) \\ (4t_\parallel, 4t_\parallel + 4t_\perp) \\ (4t_\parallel + 4t_\perp, 8t_\parallel) \\ (8t_\parallel, 8t_\parallel + 4t_\perp) \\ (8t_\parallel + 4t_\perp, +\infty) \end{cases}, \text{ shape of FS} \begin{cases} \text{non} \\ \text{ring around } \Gamma \\ \text{non} \\ \text{rings around } X, Y \\ \text{non} \\ \text{ring around } M \\ \text{non} \end{cases} . \tag{59}$$

D. Necessity of $(0,1,1,1)$ Indices

For physical consideration, $(0,1,1,1)$ index is necessary to realize side-surface states for actual current transport.

To see this, we notice the anisotropic parameter choice about x-y plane and z-direction. The Lifshitz transition picture is obtained by opening z direction. On the other hand, when we open, say x direction, the condition will be changed. The similar picture as multi-Dirac fermions will also emerges

$$H(\boldsymbol{k}) = \bigoplus_{n=1}^{L_z} \left[\lambda_\parallel \sin(k_y b)\sigma_y + \lambda_\perp \sin(k_z c)\sigma_x + m_n^x(\boldsymbol{k})\tau_z\sigma_z\right], \tag{60}$$

but with a different lowest mass

$$m_1^x(\boldsymbol{k}) = \Theta\left[-m_0^x(\boldsymbol{k})\right] m_0^x(\boldsymbol{k}) + \Theta\left[m_0^x(\boldsymbol{k}) - 4t_\perp\right](4t_\perp - m_0^x(\boldsymbol{k})), \tag{61}$$

with

$$m_1^x(\boldsymbol{k}) = m_0 - 4t_\parallel \sin^2\frac{k_y b}{2} - 4t_\perp \sin^2\frac{k_z c}{2}. \tag{62}$$

Then, accordingly, possible four Dirac point at Γ^x, Y, Z, M^x with $(k_y, k_z) = (0,0), (\pi,0), (0,\pi), (\pi,\pi)$ will emerge accordingly as

$$m_0 \in \begin{cases} (0, 4t_\parallel), & \Gamma^x \\ (4t_\parallel, 8t_\parallel), & Y \\ (4t_\perp, 4t_\parallel + 4t_\perp), & Z \\ (4t_\parallel + 4t_\perp, 8t_\parallel + 4t_\perp), & M^x \end{cases}. \tag{63}$$

The same criterion applies for y-open condition with Γ^x, Y, M^x replaced by Γ^y, X, M^y.

Essentially, notice that when $4t_\perp < m_0 < 4t_\parallel$, two gapless Dirac cones emerge at Γ and Z, which gives rise to a weak TI phase with index $(0,1,1,0)$. The SMWTI made in this case is trivial, since the windings around Γ and Z are opposite. The index here follows the definition that the first index corresponds to strong TI index, and the latter three will be when we open one direction, the top and bottom surfaces have gapless Dirac cones or not. From physical consideration, this is necessary since one needs surface on all side surfaces to bear the current. For $(0,1,1,0)$ case, there is no such surface state on z-side surface when making x-open TI film.

TABLE I. TI phases, $0 < t_\perp < t_\parallel$ assumed.

m_0	Index	$\Gamma^{x,y}$	Γ^z	X, Y	Z	$M^{x,y}$	M^z
$(-\infty, 0)$	$(0,0,0,0)$	✗	✗	✗	✗	✗	✗
$(0, 4t_\perp)$	$(1,1,1,1)$	✓	✓	✗	✗	✗	✗
$(4t_\perp, 4t_\parallel)$	$(0,1,1,0)$	✓	✗	✗	✓	✗	✗
$(4t_\parallel, 4t_\parallel + 4t_\perp)$	$(0,1,1,1)$	✗	✗	✓	✓	✗	✗
$(4t_\parallel + 4t_\perp, 8t_\parallel)$	$(0,1,1,0)$	✗	✗	✓	✗	✓	✗
$(8t_\parallel, 8t_\parallel + 4t_\perp)$	$(1,1,1,1)$	✗	✗	✗	✗	✓	✓
$(8t_\parallel + 4t_\perp, +\infty)$	$(0,0,0,0)$	✗	✗	✗	✗	✗	✗

II. $N = 4$ Phase

Simplest picture comes from a low-energy expansion

$$h_\chi^\Gamma(\boldsymbol{q}^\Gamma) = \lambda_\parallel \boldsymbol{q}^\Gamma \cdot \boldsymbol{\sigma} + \tilde{m}_\chi^\Gamma(\boldsymbol{q}^\Gamma)\sigma_z, \tag{64a}$$

$$h_\chi^X(\boldsymbol{q}^X) = -\lambda_\parallel \boldsymbol{q}^X \cdot \bar{\boldsymbol{\sigma}} + \tilde{m}_\chi^X(\boldsymbol{q}^X)\sigma_z, \tag{64b}$$

$$h_\chi^Y(\boldsymbol{q}^Y) = \lambda_\parallel \boldsymbol{q}^Y \cdot \bar{\boldsymbol{\sigma}} + \tilde{m}_\chi^Y(\boldsymbol{q}^X)\sigma_z, \tag{64c}$$

$$h_\chi^M(\boldsymbol{q}^M) = -\lambda_\parallel \boldsymbol{q}^M \cdot \boldsymbol{\sigma} + \tilde{m}_\chi^M(\boldsymbol{q}^M)\sigma_z. \tag{64d}$$

Admittedly that the in-plane handness is determined by

$$\mathcal{S} = \begin{cases} 1, \boldsymbol{\sigma} \\ -1, \bar{\boldsymbol{\sigma}} \end{cases}, \tag{65}$$

the windingness of each Dirac Fermion is given by

$$\omega = \mathcal{S} \cdot \mathrm{sgn}(\tilde{m}), \tag{66}$$

which is directly related to the topological index.

A. Absence of $N = 4$ Phase by Fermion-Doubling in SMTI

In semi-magnetic topological insulator, the masses read

$$\tilde{m}_+(\boldsymbol{q}) = V + \sqrt{V^2 + (\tilde{m}_-(\boldsymbol{q}))^2}, \tag{67a}$$

$$\tilde{m}_-(\boldsymbol{q}) = \Theta(q - q_c)(\tilde{m}_0^D - t_\parallel q^2), \tag{67b}$$

with $V = I_S = I_A \approx V_z^{\text{top}}/2 > 0$ assumed. $\tilde{m}_0^D$ means the effective static mass depends on Dirac point. Similar to case before, $\chi = +$ cones are trivial, while for Fermi surface inside the parity invariant regime, the Hall conductance becomes

$$\sigma_H = \frac{e^2}{2h}\omega = -\frac{e^2}{2h}\mathcal{S}, \tag{68}$$

which is negatively half-quantized for Γ and M, while the opposite for X, Y, and they cancel each other totally.

The deep reason lies behind is the Fermion doubling theorem[18, 19]. By counting $N = 4$ in a single band we have included all the Dirac fermions formed by $\boldsymbol{k} \cdot \boldsymbol{\sigma}$ on lattice, which introduces equal number of Dirac fermions with opposite 'handness'. Since the Hall conductance is directly proportional to this 'handness' and the high-energy mass sign, which is provided to be the same by the static Zeeman splitting, the total Hall conductance must be zero.

B. $N = 2 + 2$ Phase by Staggered Potential

A staggered potential saves the condition. Introduce $V(\boldsymbol{k})$ to replace V above, with requirement $V(G/M) = -V(X/Y)$, or a sub-anti-periodic condition as

$$V(k_x + \pi, k_y) = -V(k_x, k_y) = V(k_x, k_y + \pi), \tag{69}$$

with $V(0, 0) > 0$, we immediately find that the four gapless Dirac cones are assigned to $\chi = +$ band at X, Y, and $\chi = -$ band at Γ, G, with each band contributing $-e^2/h$ Hall conductance and totally $-Ne^2/2h$ with $N = 4 = 2 + 2$. Simple picture comes from separate Dirac cones as

$$\sigma_H^{\Gamma,M} = \frac{e^2}{2h}\mathcal{S}(\sigma) \cdot \{-\text{sgn}[V(G/M)]\} = -\frac{e^2}{2h}, \tag{70a}$$

$$\sigma_H^{X,Y} = \frac{e^2}{2h}\mathcal{S}(\bar{\sigma}) \cdot \{-\text{sgn}[V(X/Y)]\} = -\frac{e^2}{2h}, \tag{70b}$$

since by observation we have that for χ band to be non-trivial,

$$\text{sgn}(\tilde{m}_\chi(q = +\infty)) = -\text{sgn}V(q = 0), \tag{71}$$

as a well-established correspondence.

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